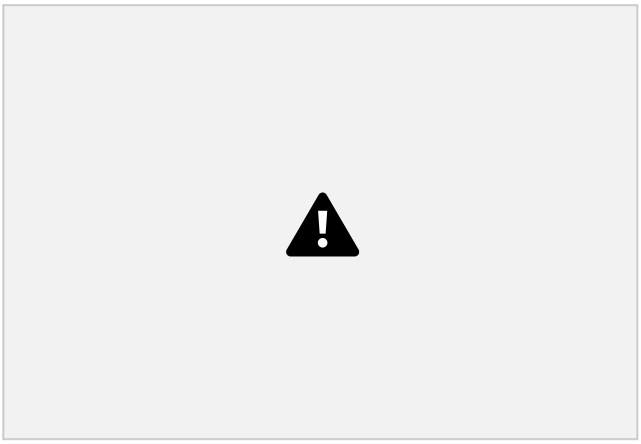
DSA06 - Data Handling and Visualization | Activity 5: Normality Detection using Q-Q Plots | Slot D

R.No: 32 Reg. No: 1924260143 Name: S.Gopikha

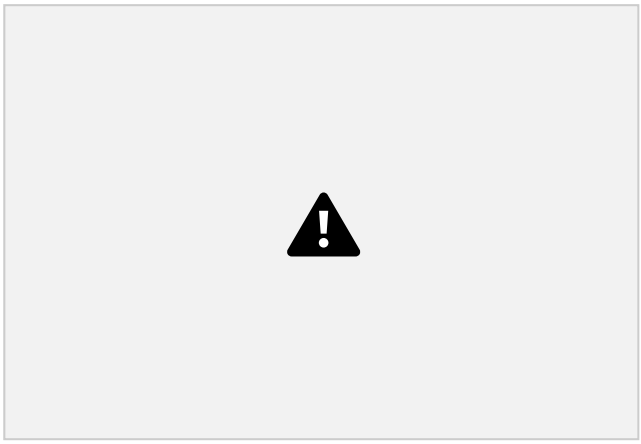
Each chart is a normal Q-Q plot made in R (stat_qq and stat_qq_line; red line = perfect normal fit). Part A (16 marks): for every chart decide whether the data are normal and name the shape from this list: Normal, Right-skewed, Left-skewed, Heavy-tailed, Light-tailed, Bimodal, Normal with outlier(s); then cite the evidence you see.

Chart 1: lab experiment duration (minutes), n = 40



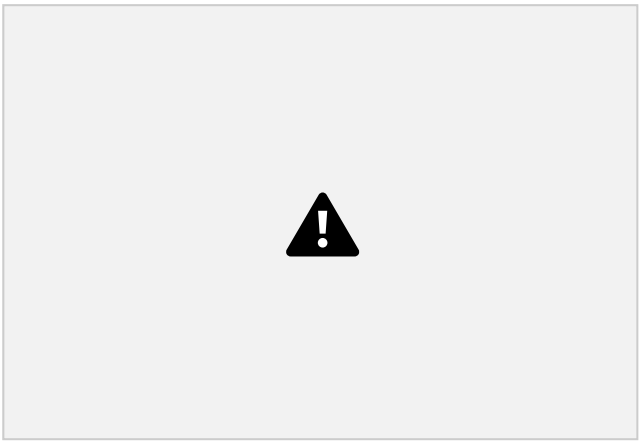
Normal? (Yes / No)	
Shape (from the list)	
Evidence in the plot	

Chart 2: typing speed (words per minute), n = 50



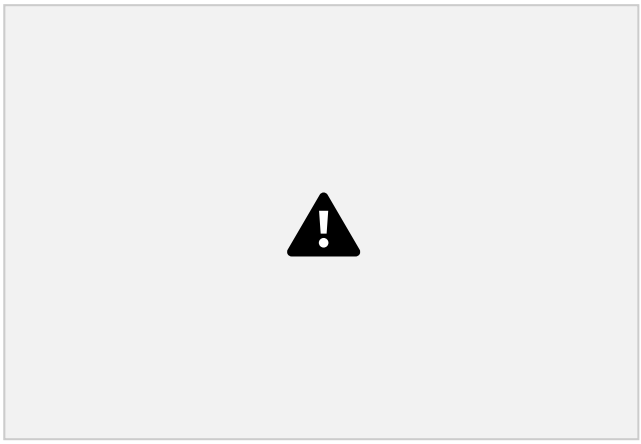
Normal? (Yes / No)	
Shape (from the list)	
Evidence in the plot	

Chart 3: lab experiment duration (minutes), n = 50



Normal? (Yes / No)	

Chart 4: plant height (cm), n = 80



Shape (from the list)	
Evidence in the plot	

Evidence in the plot	

Part B (4 marks)

Normal? (Yes / No)	
Shape (from the list)	

B1. Chart 4 departs from normality. Name the departure and state the remedy you would try before using mean-and-SD methods (for example log transform, square, split the data, robust methods). [2]

B2. Fill the blanks to draw a Q-Q plot with its reference line for a data frame df with a column value: ggplot(df, aes(sample = value)) + _____ + _____ [2]

Part A: Q-Q Plot Analysis

Fill in the answers based on the plots:

Chart	Normal?	Shape	Evidence in the plot
Chart 1	Check the plot	Choose from the given list	Describe how the points follow or deviate from the red line.
Chart 2	Check the plot	Choose from the given list	Describe the curvature or tail deviations from the red line.
Chart 3	Check the plot	Choose from the given list	Identify whether the points form a curve, an S-shape, or separate groups.
Chart 4	No (given in B1)	Check the plot	Describe the visible departure from the red reference line.

How to identify the shapes

- **Normal:** Points lie approximately along the red line.
- **Right-skewed:** Points show a curved pattern associated with a longer right tail.
- **Left-skewed:** Points show a curved pattern associated with a longer left tail.
- **Heavy-tailed:** Both tails deviate outward from the reference line.
- **Light-tailed:** Both tails deviate inward toward the reference line.
- **Bimodal:** Points show a distinctive pattern suggesting two groups or peaks.
- **Normal with outlier(s):** Most points follow the line, but one or a few points lie far away from it.

Part B:

B1. Departure from normality and remedy

Since Chart 4 departs from normality, identify its specific departure by examining the plot.

Answer format:

- **Departure:** [Right-skewed / Left-skewed / Heavy-tailed / Light-tailed / Bimodal / Normal with outlier(s)]
- **Remedy:** Apply an appropriate transformation, such as a log transformation for right-skewed positive data. If the data contain distinct groups, consider analyzing the groups separately. Robust statistical methods may be appropriate when outliers are present.

B2. R code for a Q-Q plot

Fill in the blanks as follows:

```
ggplot(df, aes(sample = value)) +
  stat_qq() +
  stat_qq_line()
```

- `stat_qq()` creates the normal Q-Q plot.
- `stat_qq_line()` adds the reference line.

Please upload a photo or screenshot showing Charts 1–4. I'll then give you the exact Normal? answer, shape, and plot evidence for each chart, along with the specific answer for B1.